



(12) **United States Patent**
Nakatsuka et al.

(10) **Patent No.:** **US 12,416,360 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **MECHANICAL SEAL DEVICE**

(56) **References Cited**

(71) Applicant: **PILLAR CORPORATION**, Osaka
(JP)

U.S. PATENT DOCUMENTS

(72) Inventors: **Masaya Nakatsuka**, Osaka (JP);
Masashi Hayashi, Osaka (JP);
Shigeyuki Fujinaga, Osaka (JP)

2,628,852 A * 2/1953 Voytech F16J 15/38
165/47
3,495,840 A * 2/1970 Wilk F16J 15/342
277/408

(Continued)

(73) Assignee: **PILLAR CORPORATION**, Osaka
(JP)

FOREIGN PATENT DOCUMENTS

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

CN 105649983 A 6/2016
JP 2001-059487 A 3/2001

(Continued)

OTHER PUBLICATIONS

(21) Appl. No.: **18/856,308**

International Search Report issued in corresponding International
Application No. PCT/JP2023/000288, dated Mar. 28, 2023 (5
pages).

(22) PCT Filed: **Jan. 10, 2023**

(86) PCT No.: **PCT/JP2023/000288**

§ 371 (c)(1),

(2) Date: **Oct. 11, 2024**

Primary Examiner — Nicholas L Foster

(74) *Attorney, Agent, or Firm* — Osha Bergman Watanabe
& Burton LLP

(87) PCT Pub. No.: **WO2023/243127**

PCT Pub. Date: **Dec. 21, 2023**

(57) **ABSTRACT**

(65) **Prior Publication Data**

US 2025/0189040 A1 Jun. 12, 2025

A circulation pump mechanism of a mechanical seal device includes a rotor having mounting grooves formed therein, vanes provided in the mounting grooves, a plurality of biasing members provided in a space on a deep side with respect to the vane in each mounting groove and configured to bias each vane in a direction in which the vane is caused to project, a housing having an inner circumferential surface which is eccentric with respect to an outer circumferential surface of the rotor and on which a tip of each vane slides, a pump chamber demarcated on a leading side in a rotation direction of the rotor with respect to the tip of each vane between the outer circumferential surface of the rotor and the inner circumferential surface of the housing, and a communication flow path formed in the rotor and providing communication between the space and the pump chamber.

(30) **Foreign Application Priority Data**

Jun. 16, 2022 (JP) 2022-096960

(51) **Int. Cl.**

F16J 15/34 (2006.01)

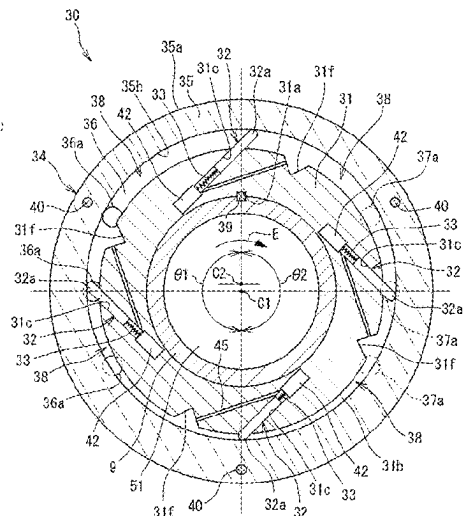
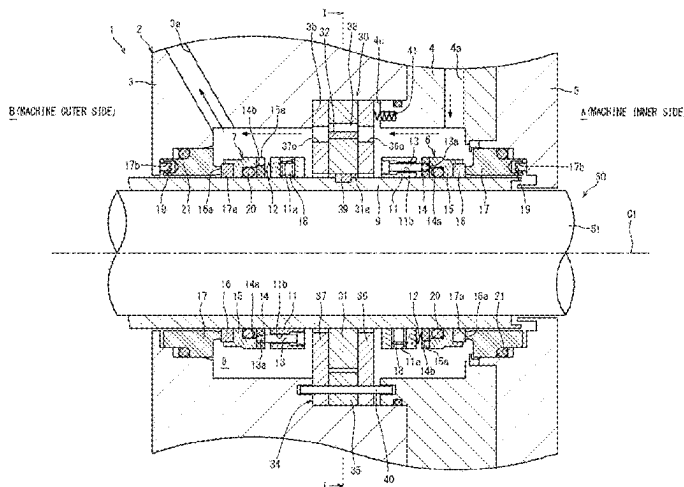
(52) **U.S. Cl.**

CPC **F16J 15/3404** (2013.01); **F16J 15/3484**
(2013.01)

(58) **Field of Classification Search**

CPC F16J 15/3404; F16J 15/3484; F04C 2/344
See application file for complete search history.

5 Claims, 11 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

3,746,350	A *	7/1973	Mayer		F16J 15/3404	277/408
3,888,495	A *	6/1975	Mayer		F04D 29/043	277/927
4,545,588	A *	10/1985	Nagai		F16J 15/346	277/408
4,721,311	A *	1/1988	Kakabaker		F16J 15/162	277/408
5,195,867	A *	3/1993	Stirling		F16J 15/3404	415/111
9,347,565	B2 *	5/2016	Berger		F16J 15/3404	
11,441,571	B2 *	9/2022	Yoshikawa		F04D 29/128	
11,698,080	B2 *	7/2023	Yoshikawa		F04D 29/124	415/111

FOREIGN PATENT DOCUMENTS

JP	2006-144682	A	6/2006
JP	2022-018073	A	1/2022

* cited by examiner

FIG. 1

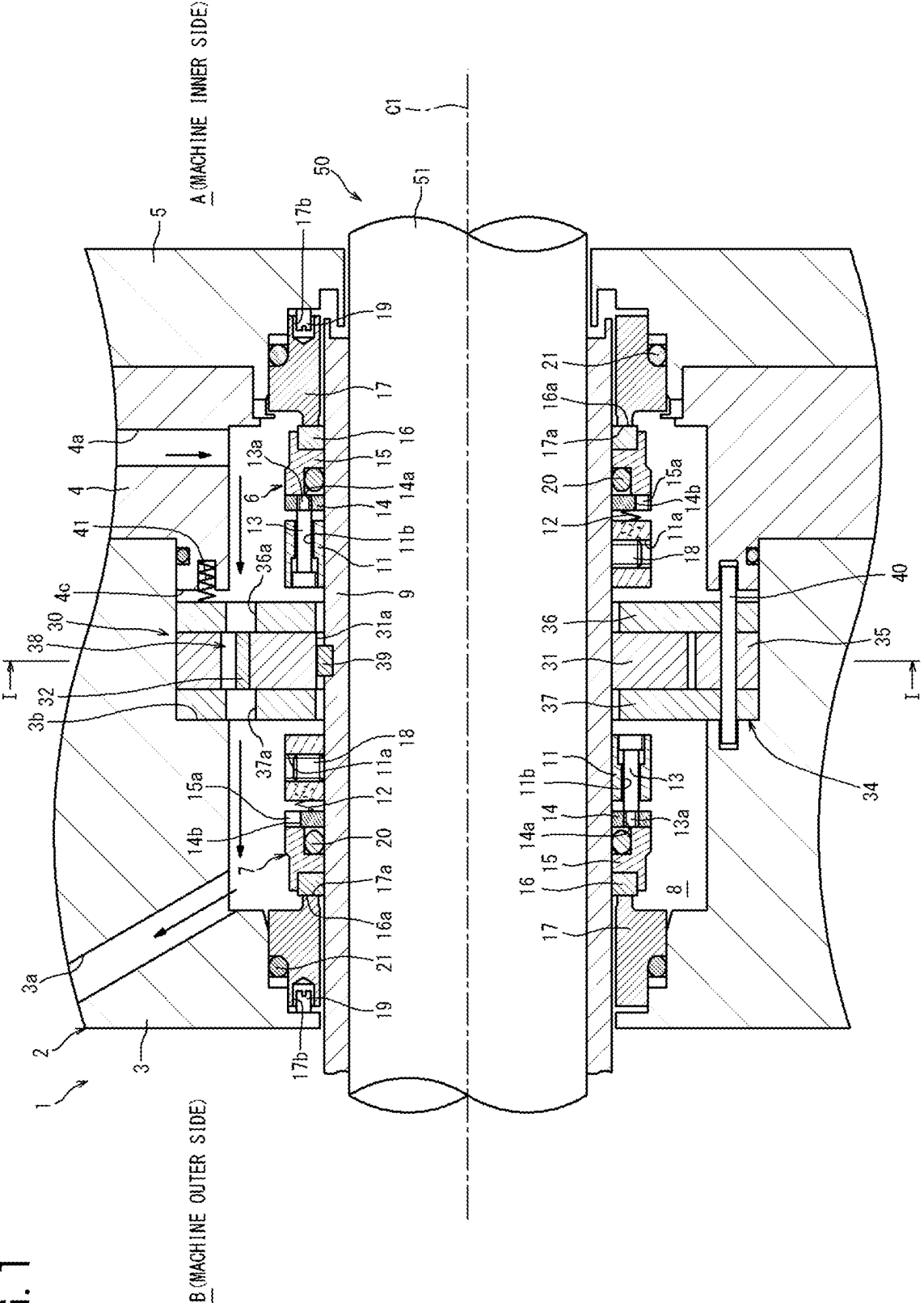


FIG. 2

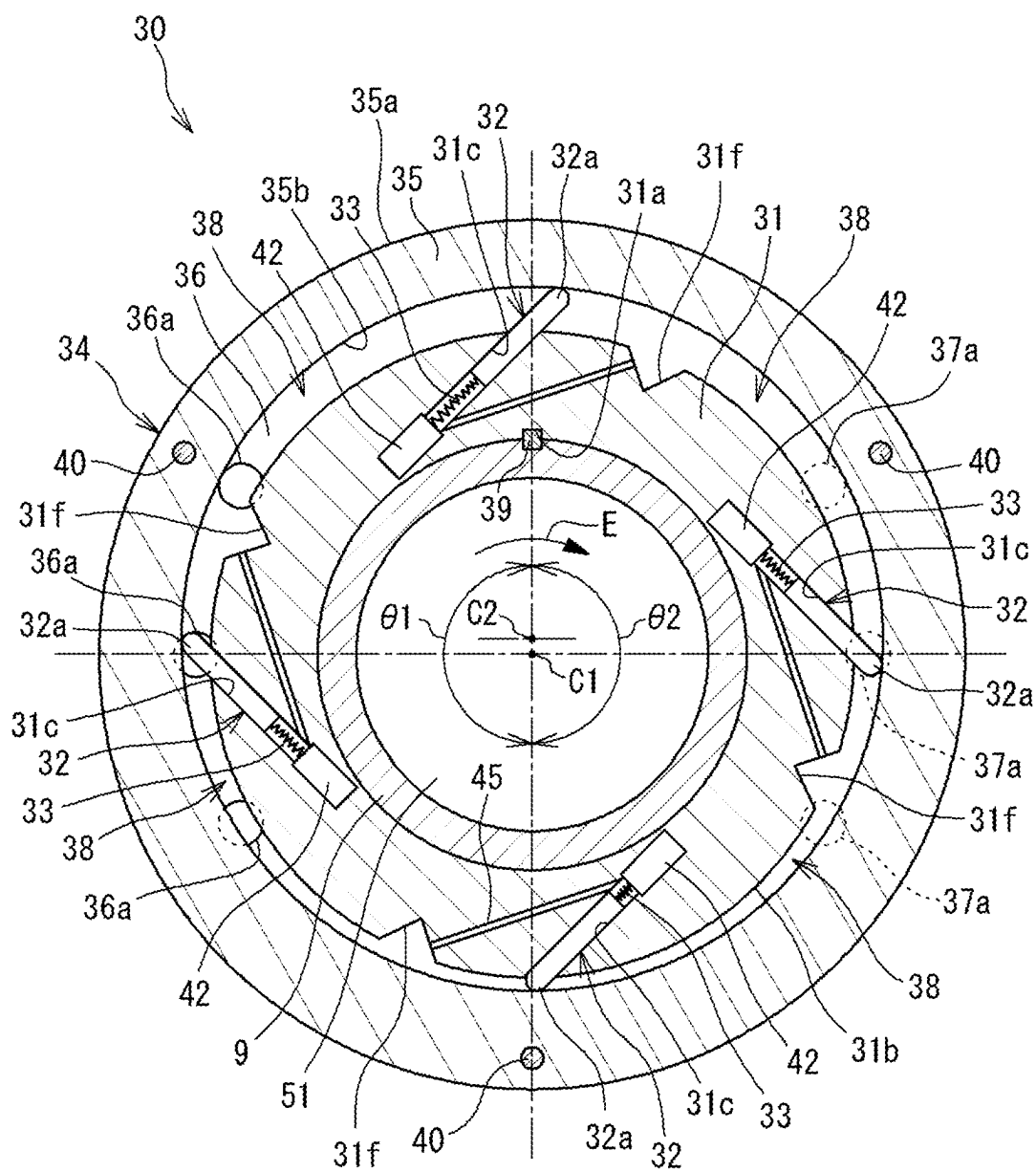


FIG. 3

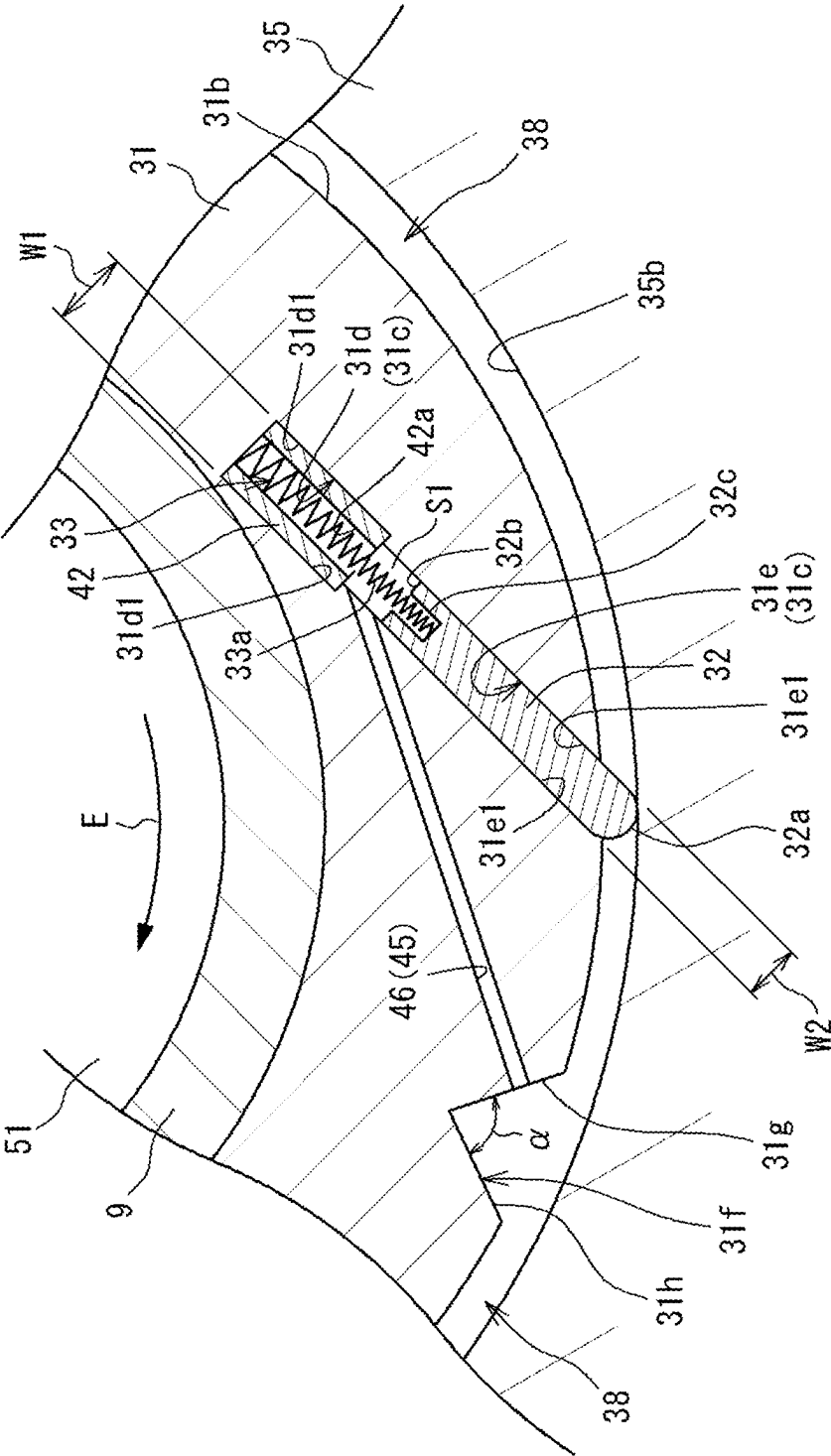


FIG. 4

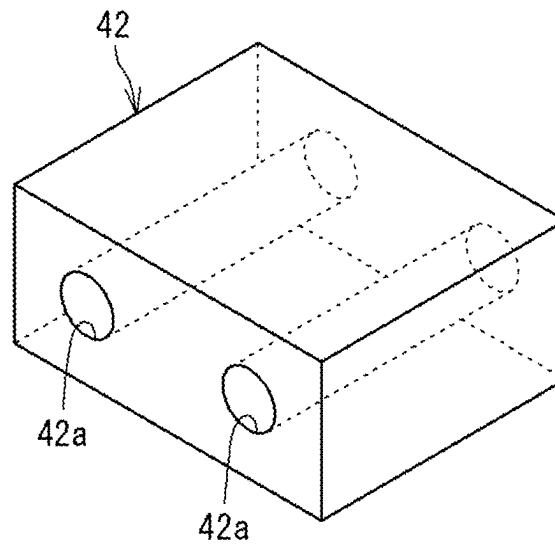


FIG. 5

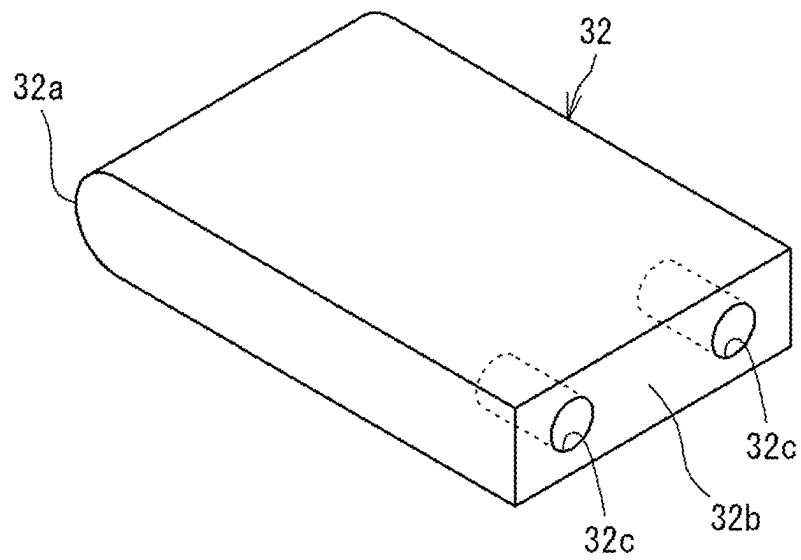


FIG. 6

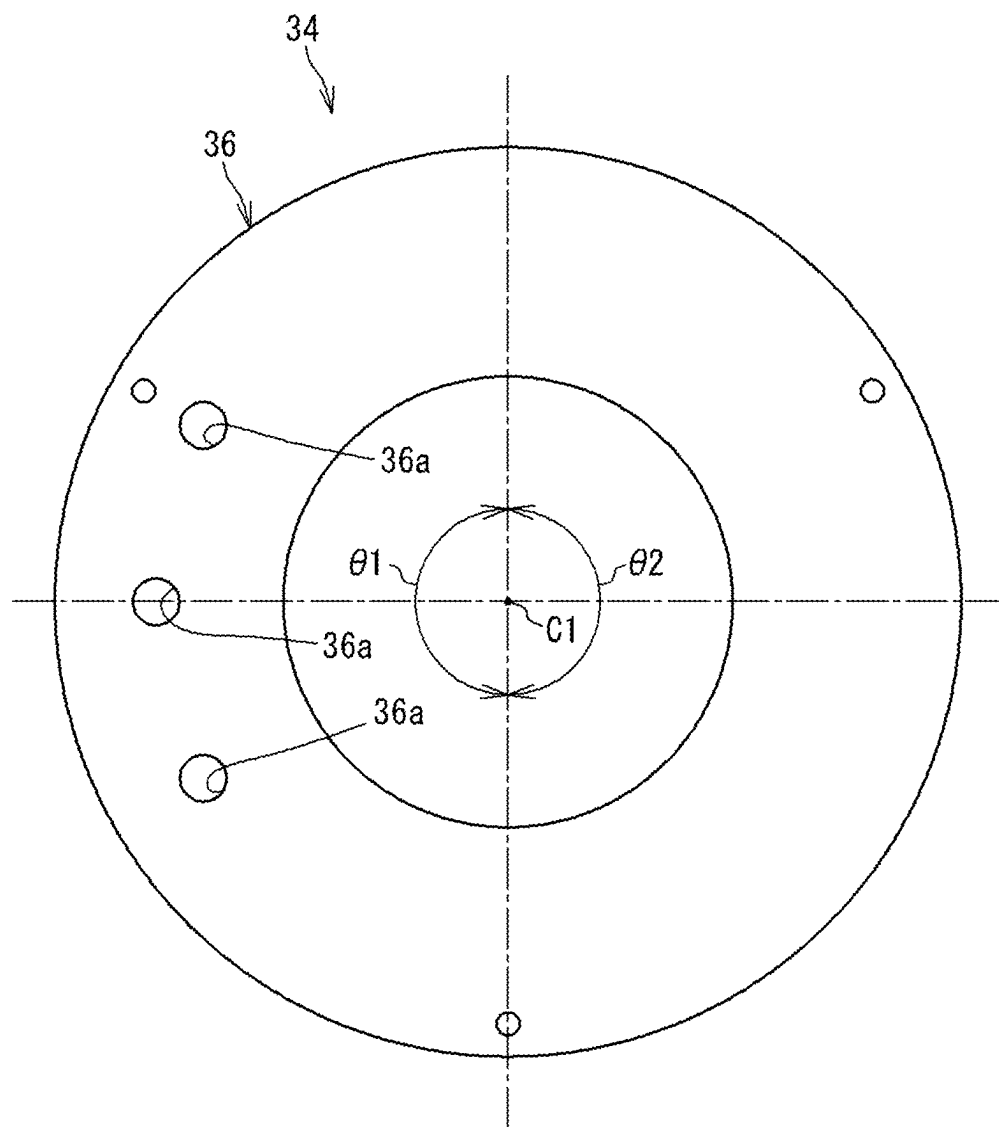


FIG. 7

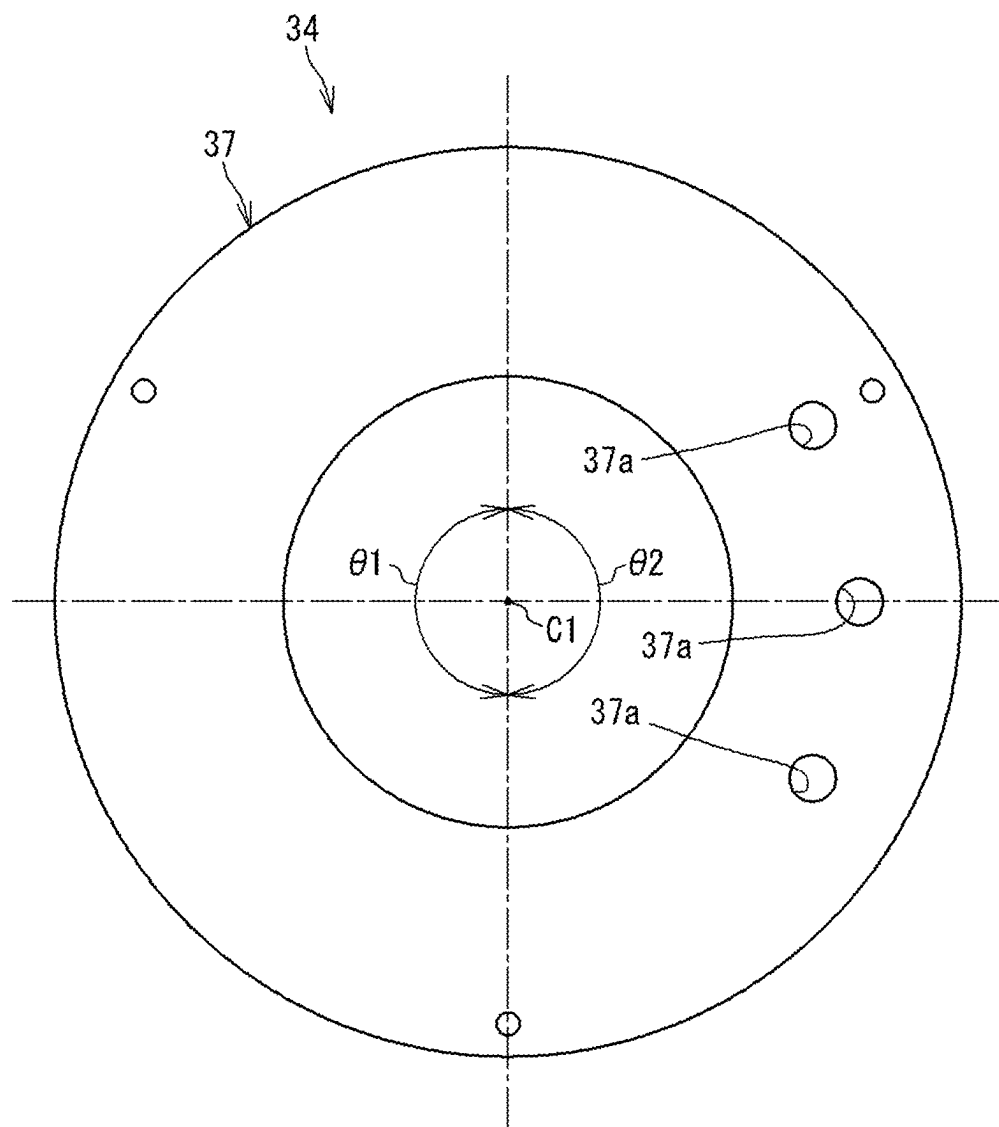


FIG. 8

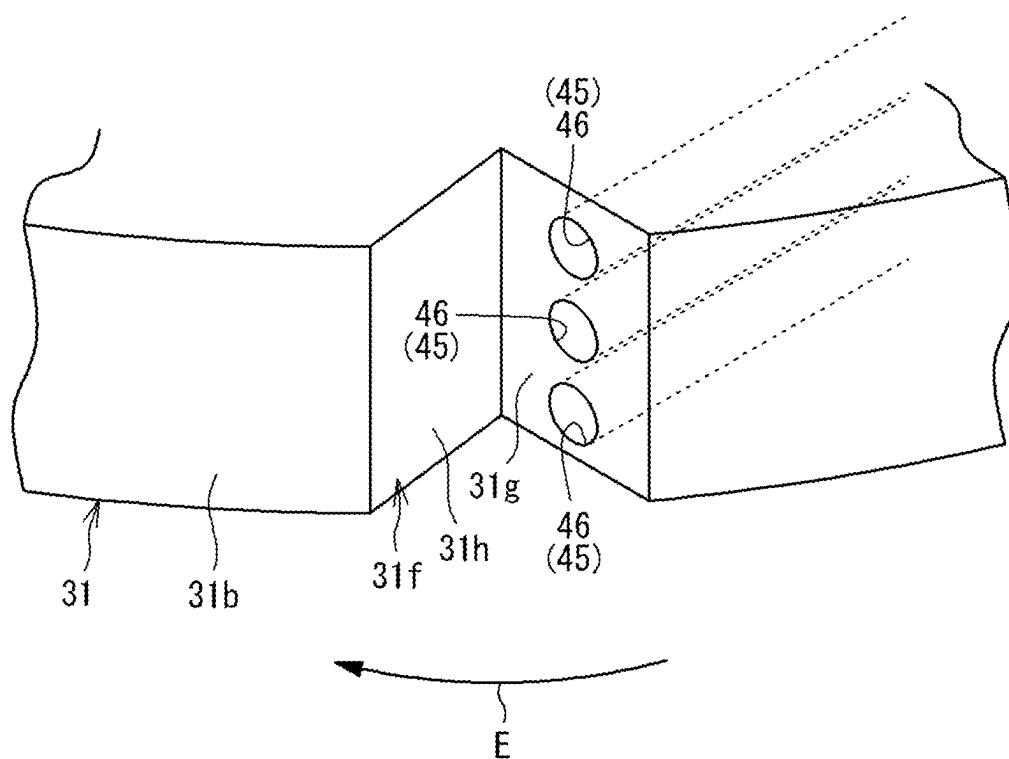


FIG. 9

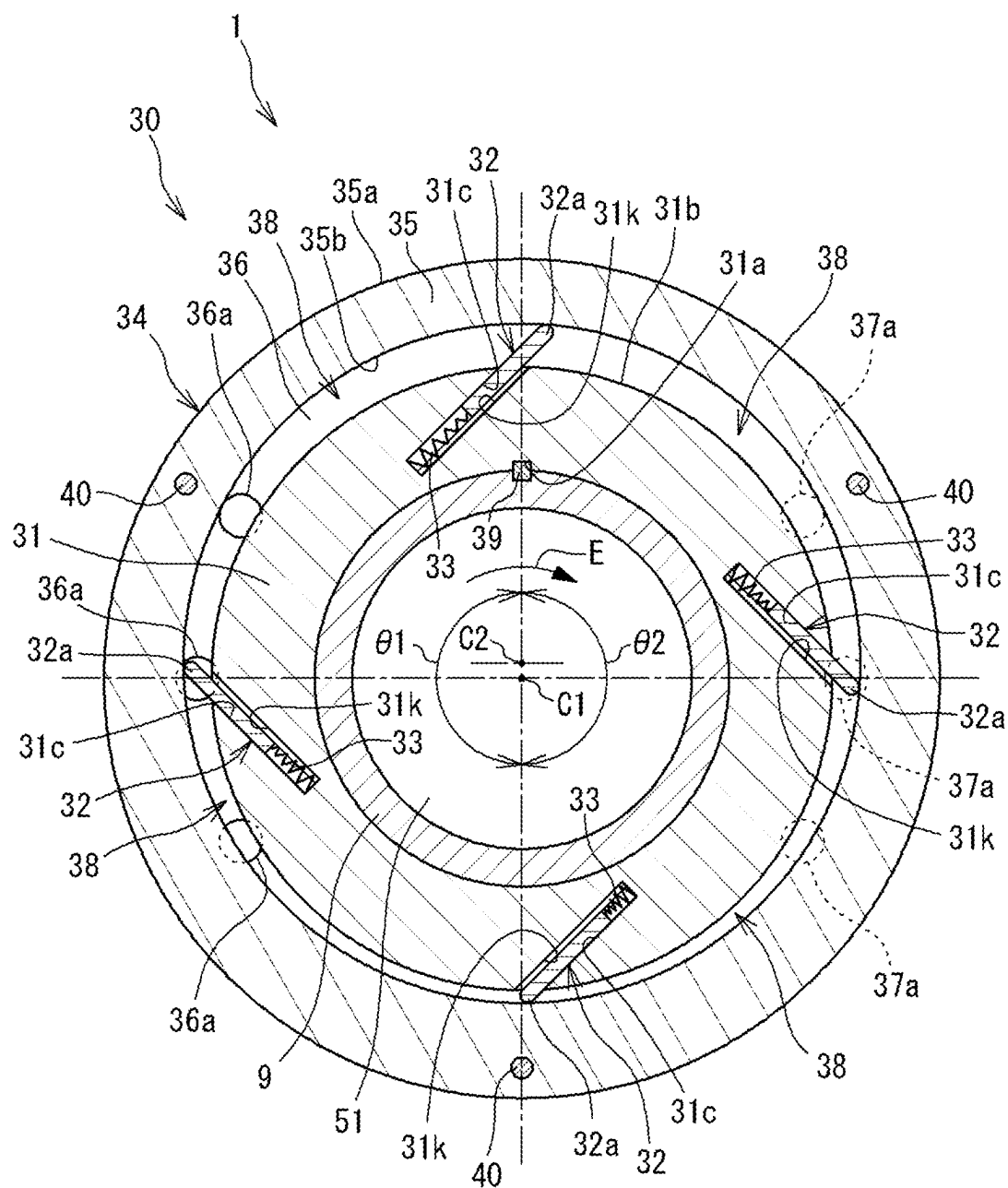


FIG. 10

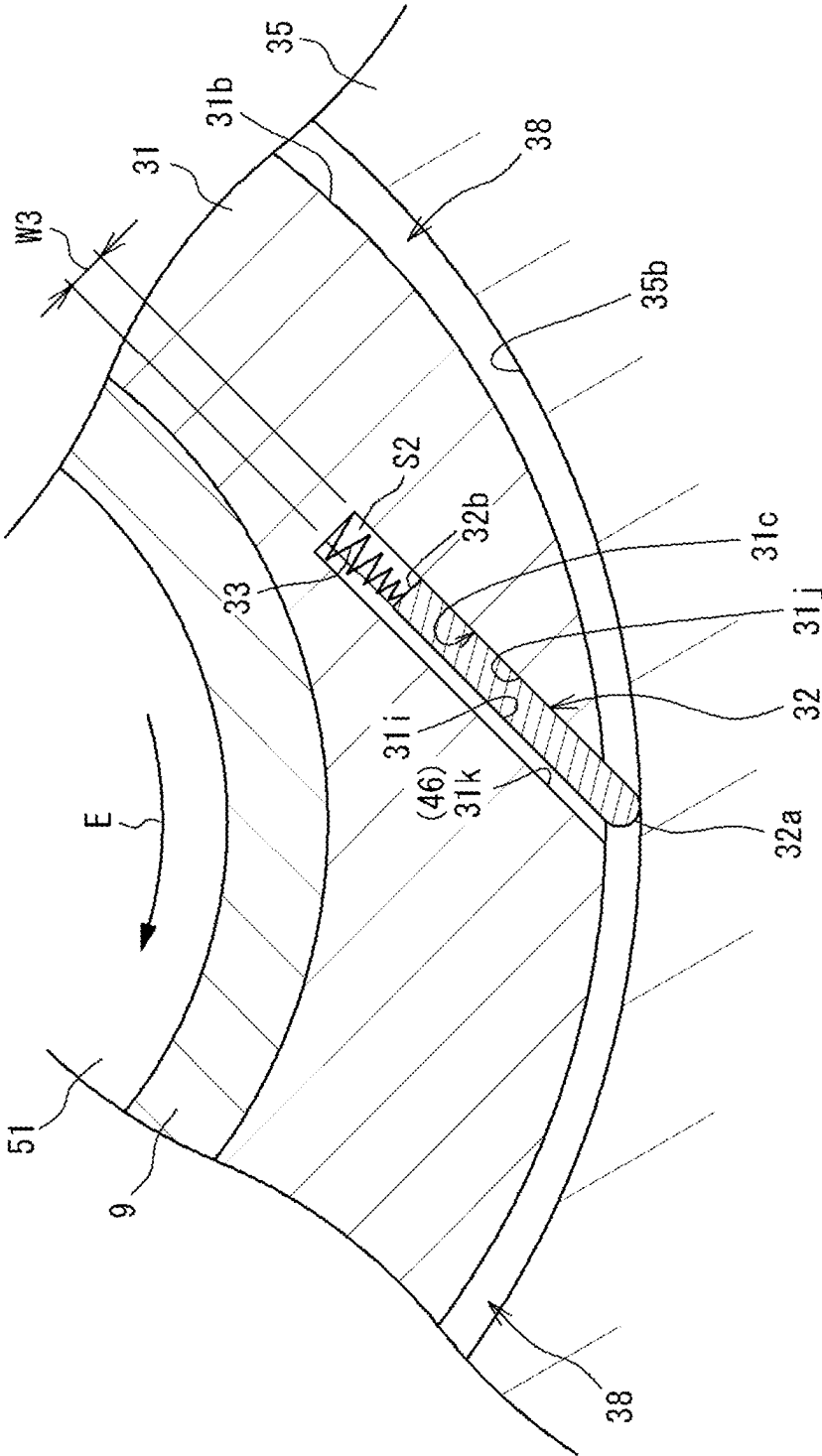
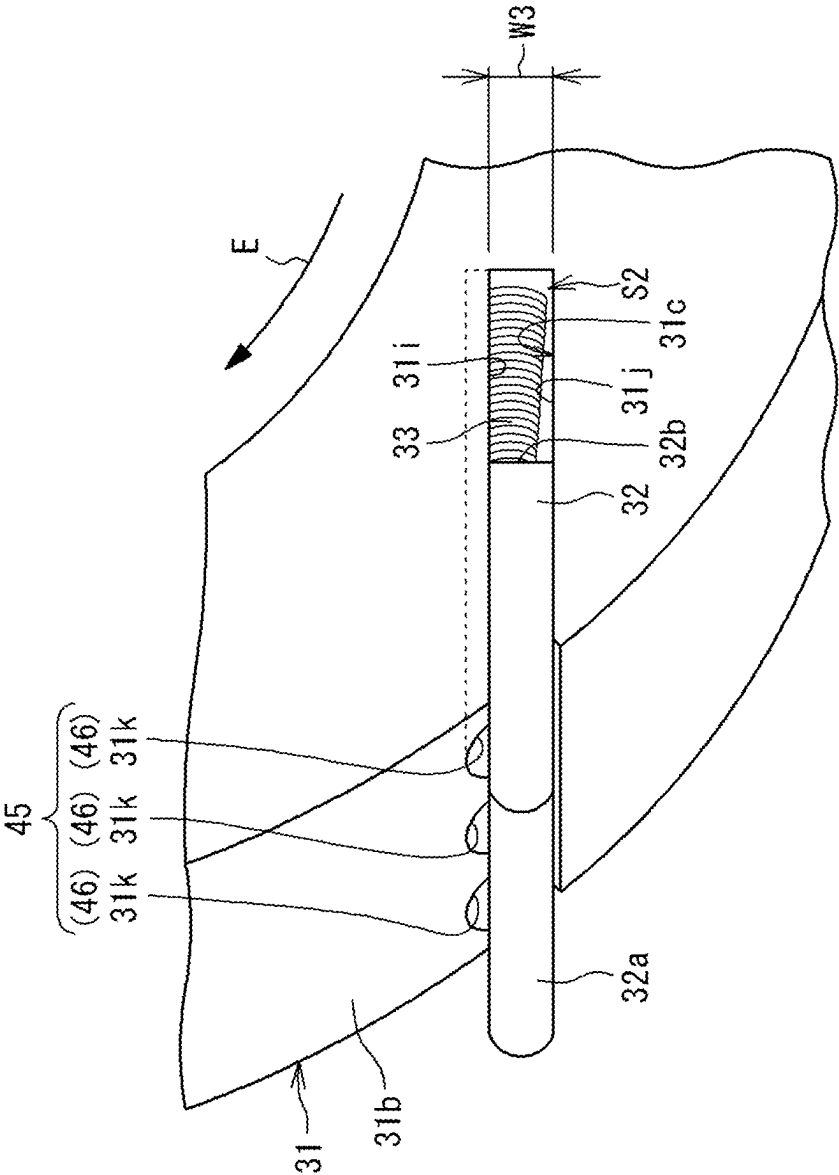


FIG. 11



1

MECHANICAL SEAL DEVICE**TECHNICAL FIELD**

The present invention relates to a mechanical seal device. 5

BACKGROUND ART

A mechanical seal device installed in a rotary machine includes a circulation pump mechanism that circulates a sealing liquid in order to enhance lubrication between sealing surfaces of a rotary sealing ring and a fixed sealing ring and to remove sliding heat generated between the sealing surfaces (see, for example, PATENT LITERATURE 1). A mechanical seal device of PATENT LITERATURE 1 includes a circulation pump mechanism that functions as a vane pump. The circulation pump mechanism includes a rotor that rotates integrally with a rotary shaft of a rotary machine, a plurality of vanes that freely project and retract from the outer circumferential surface of the rotor, springs that respectively bias the vanes in a direction in which the vanes are caused to project, and a plurality of pump chambers that are each demarcated between the adjacent vanes. Such a circulation pump mechanism can circulate a fixed volume of a sealing liquid each time the rotor makes one rotation, and thus can circulate the required flow volume of the sealing liquid even in a rotary machine that rotates at low speeds, such as an agitator.

CITATION LIST

Patent Literature

PATENT LITERATURE 1: Japanese Laid-Open Patent Publication No. 2022-18073

SUMMARY OF THE INVENTION**Technical Problem**

In the circulation pump mechanism of PATENT LITERATURE 1, the tip of each vane rotating together with the rotor slides on the inner circumferential surface of the housing which is eccentric with respect to the outer circumferential surface of the rotor. Accordingly, when the pressing force from the inner circumferential surface of the housing becomes stronger due to the rotation of the rotor, each vane retracts into the rotor against the biasing force of the spring. When the pressing force from the inner circumferential surface of the housing becomes weaker due to the rotation of the rotor, each vane having retracted into the rotor projects from the outer circumferential surface of the rotor by the biasing force of the spring. However, the pressure of the sealing liquid in the pump chamber also acts on the tip of each vane having retracted into the rotor. Therefore, if the pressure of the sealing liquid increases, there is a concern that each vane having retracted into the rotor does not project from the outer circumferential surface of the rotor.

The present disclosure has been made in view of such circumstances, and an object of the present disclosure is to provide a mechanical seal device that can suppress interference with projection movement of a vane with respect to a rotor in a circulation pump mechanism.

Solution to Problem

(1) The present disclosure is directed to a mechanical seal device including: a seal case surrounding a rotary shaft of a

2

rotary machine and separating a machine inner area and a machine outer area of the rotary machine from each other; at least one mechanical seal placed between the seal case and the rotary shaft; a sealing liquid chamber which is formed in the seal case and into which a sealing liquid for lubricating a sliding portion of the mechanical seal is introduced; and a circulation pump mechanism configured to circulate the sealing liquid in the sealing liquid chamber, wherein the circulation pump mechanism includes an annular rotor provided on the rotary shaft side so as to be integrally rotatable with the rotary shaft and having a mounting groove open at an outer circumferential surface thereof, a vane provided in the mounting groove and configured to freely project and retract with respect to the outer circumferential surface of the rotor, a biasing member provided in a space on a deep side with respect to the vane in the mounting groove and configured to bias the vane in a direction in which the vane is caused to project from the outer circumferential surface of the rotor, a housing having, on an inner circumference thereof, a sliding surface which is eccentric with respect to the outer circumferential surface of the rotor and on which a tip of the vane slides as the rotor rotates, the housing being mounted to the seal case, a pump chamber which is demarcated on a leading side in a rotation direction of the rotor with respect to the tip of the vane between the outer circumferential surface of the rotor and the sliding surface of the housing and into and from which the sealing liquid is sucked and discharged, and a communication flow path formed in the rotor and providing communication between the space of the mounting groove and the pump chamber.

In the mechanical seal device of the present disclosure, the space on the deep side with respect to the vane in the mounting groove of the rotor of the circulation pump mechanism communicates with the pump chamber demarcated on the leading side in the rotation direction of the rotor with respect to the tip of the vane, via the communication flow path. Therefore, the sealing liquid in the pump chamber flows into the space of the mounting groove via the communication flow path. Accordingly, the pressure of the sealing liquid in the space of the mounting groove and the pressure of the sealing liquid in the pump chamber cancel each other, so that substantially only the biasing force of the spring acts on the vane. Therefore, even if the pressure of the sealing liquid in the pump chamber acting on the tip of the vane increases, interference with the projection movement of the vane with respect to the rotor can be suppressed.

(2) In the mechanical seal device of (1) above, preferably, the communication flow path has a plurality of flow path portions aligned in an axial direction of the rotor.

In this case, the sealing liquid in the pump chamber can be quickly caused to flow into the space of the mounting groove via the plurality of flow path portions.

(3) In the mechanical seal device of (1) or (2) above, preferably, the communication flow path is formed in the rotor independently of the mounting groove.

In this case, when the sealing liquid flows in the communication flow path from the pump chamber toward the space of the mounting groove, the flow of the sealing liquid does not interfere with the projection movement of the vane. Accordingly, interference with the projection movement of the vane with respect to the rotor can be further inhibited.

(4) In the mechanical seal device of (3) above, preferably, a cut groove having a V-shape in an axial view is formed at a location, facing the pump chamber, on the outer circumferential surface of the rotor, and the communication flow path is formed so as to be open at a side surface of the cut groove.

3

In this case, the communication flow path can be formed more easily than in the case where the communication flow path is formed so as to be open at the outer circumferential surface of the rotor.

(5) In the mechanical seal device of (3) or (4) above, preferably, the communication flow path has a flow path portion composed of a hole having a circular cross-section.

In this case, the communication flow path can be formed even more easily.

Advantageous Effects of the Invention

The mechanical seal device of the present disclosure can suppress interference with the projection movement of the vane with respect to the rotor in the circulation pump mechanism.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a cross-sectional view showing a mechanical seal device according to a first embodiment of the present disclosure.

FIG. 2 is a cross-sectional view as seen in the direction of arrows I-I in FIG. 1.

FIG. 3 is an enlarged cross-sectional view showing an area around 6 o'clock in FIG. 2 in a circulation pump mechanism.

FIG. 4 is a perspective view showing a retaining member.

FIG. 5 is a perspective view showing a vane.

FIG. 6 is a side view of a first outer side plate of a housing as seen from a machine outer side.

FIG. 7 is a side view of a second outer side plate of the housing as seen from the machine outer side.

FIG. 8 is an enlarged perspective view showing an area around a cut groove of a rotor.

FIG. 9 is a cross-sectional view of a circulation pump mechanism of a mechanical seal device according to a second embodiment of the present disclosure.

FIG. 10 is an enlarged cross-sectional view showing an area around 6 o'clock in FIG. 9 in the circulation pump mechanism of the second embodiment.

FIG. 11 is an enlarged perspective view showing an area around a mounting groove of a rotor of the second embodiment.

DETAILED DESCRIPTION

Next, preferred embodiments of the present disclosure will be described with reference to the accompanying drawings. At least parts of the embodiments described below may be combined as desired.

First Embodiment

[Mechanical Seal Device]

FIG. 1 is a cross-sectional view showing a mechanical seal device 1 according to a first embodiment of the present disclosure. The mechanical seal device 1 is provided in a rotary machine 50 such as a pump and an agitator that handles a sealed fluid. The mechanical seal device 1 includes a seal case 2, a first mechanical seal 6, a second mechanical seal 7, a sealing liquid chamber 8, and a circulation pump mechanism 30.

The seal case 2 is fixed to a casing (not shown) of the rotary machine 50, and separate a machine inner area A and a machine outer area B of the rotary machine 50 from each other. The seal case 2 of the present embodiment has a first

4

case body 3, a second case body 4, and a third case body 5 in order from the machine outer area B side toward the machine inner area A side. The first case body 3, the second case body 4, and the third case body 5 are each formed in an annular shape and surround a rotary shaft 51 of the rotary machine 50. Hereinafter, in the present specification, the machine inner area A side is referred to as machine inner side, and the machine outer area B side is referred to as machine outer side.

The first mechanical seal 6 and the second mechanical seal 7 constitute a double-type mechanical seal and are placed between the seal case 2 and the rotary shaft 51 so as to be spaced at a predetermined interval in the axial direction of the rotary shaft 51 (hereinafter referred to simply as "axial direction"). The first mechanical seal 6 is placed on the machine inner side and seals between the machine inner area A and the sealing liquid chamber 8 (described later). The second mechanical seal 7 is placed on the machine outer side and seals between the machine outer area B and the sealing liquid chamber 8.

The first mechanical seal 6 includes a spring retainer 11, springs 12, drive pins 13, a drive collar 14, a retainer 15, a rotary sealing ring 16, a stationary sealing ring 17, set screws 18, pins 19, an O-ring 20, and an O-ring 21.

The spring retainer 11 is formed in an annular shape and is fixed via a sleeve 9 on the outer circumferential side of the rotary shaft 51. A plurality of screw holes 11a (only one is shown in FIG. 1) penetrating in the radial direction are formed in the spring retainer 11 so as to be spaced at predetermined intervals in the circumferential direction. The set screw 18 is screwed into each screw hole 11a, and the tip of the set screw 18 presses the outer circumferential surface of the sleeve 9. Accordingly, the spring retainer 11 is fixed to the rotary shaft 51 side by the set screws 18.

A plurality of through holes 11b (only one is shown in FIG. 1) penetrating in the axial direction are formed in the spring retainer 11 so as to be spaced at predetermined intervals in the circumferential direction. The drive pin 13 is inserted into each through hole 11b and is held so as to be movable within a predetermined range in the axial direction relative to the spring retainer 11.

An end portion on the machine inner side of each drive pin 13 is fixed to the annular drive collar 14 which is mounted on the outer circumferential surface of the sleeve 9 on the machine inner side with respect to the spring retainer 11 so as to be movable in the axial direction. Specifically, a plurality of screw holes 14a (only one is shown in FIG. 1) penetrating in the axial direction are formed in the drive collar 14 so as to be spaced at predetermined intervals in the circumferential direction. An externally threaded portion 13a formed on the end portion on the machine inner side of each drive pin 13 is screwed into each screw hole 14a. Accordingly, the drive collar 14 is restricted from rotating relative to the spring retainer 11 while being held via the drive pin 13 so as to be movable in the axial direction relative to the spring retainer 11.

The spring retainer 11 holds end portions of a plurality of the springs 12 (only one is shown in FIG. 1) at predetermined intervals in the circumferential direction. Other end portions of the springs 12 are in contact with the end surface on the machine outer side of the drive collar 14. Accordingly, the drive collar 14 is biased toward the machine inner side by the biasing forces of the springs 12.

The annular retainer 15 is placed adjacent to the drive collar 14 on the machine inner side with respect to the drive collar 14. The retainer 15 is mounted on the outer circumferential surface of the sleeve 9 so as to be movable in the

5

axial direction. A plurality of engagement pins **15a** (only one is shown in FIG. 1) projecting toward the machine outer side are provided on the end surface on the machine outer side of the retainer **15** so as to be spaced at predetermined intervals in the circumferential direction.

Each engagement pin **15a** is engaged with an engagement hole **14b** formed in the drive collar **14**. Accordingly, the retainer **15** is restricted from rotating relative to the drive collar **14** while being held together with the drive collar **14** so as to be movable in the axial direction relative to the rotary shaft **51**. The O-ring **20** made of rubber is provided on the inner circumference of the retainer **15** so as to seal (secondarily seal) between the outer circumferential surface of the sleeve **9** and the inner circumference of the retainer **15**.

The annular rotary sealing ring **16** is fitted and fixed on the machine inner side of the retainer **15**. The end surface on the machine inner side of the rotary sealing ring **16** is a sealing surface **16a**. The stationary sealing ring **17** is placed adjacent to the rotary sealing ring **16** on the machine inner side with respect to the rotary sealing ring **16**. The stationary sealing ring **17** is fitted and fixed to the inner circumferential side of the third case body **5**. The end surface on the machine outer side of the stationary sealing ring **17** is a sealing surface **17a** on which the sealing surface **16a** of the rotary sealing ring **16** slides.

A plurality of pin grooves **17b** (only one is shown in FIG. 1) are formed on the machine inner side of the stationary sealing ring **17** so as to be spaced at predetermined intervals in the circumferential direction. The pin **19** which is fixed on the inner circumferential side of the third case body **5** so as to project toward the machine outer side is inserted into each pin groove **17b**. Accordingly, the stationary sealing ring **17** is restricted from rotating relative to the third case body **5**. The O-ring **21** made of rubber is provided on the outer circumference of the stationary sealing ring **17** so as to seal (secondarily seal) between the inner circumferential surface of the third case body **5** and the outer circumference of the stationary sealing ring **17**.

Similar to the first mechanical seal **6**, the second mechanical seal **7** includes a spring retainer **11**, springs **12**, drive pins **13**, a drive collar **14**, a retainer **15**, a rotary sealing ring **16**, a stationary sealing ring **17**, set screws **18**, pins **19**, an O-ring **20**, and an O-ring **21**. The spring retainer **11** to the O-ring **21** that are the components of the second mechanical seal **7** are arranged so as to be symmetrical with the spring retainer **11** to the O-ring **21** that are the components of the first mechanical seal **6** in the axial direction with the circulation pump mechanism **30** interposed therebetween.

Each pin **19** of the second mechanical seal **7** is fixed on the inner circumferential side of the first case body **3** so as to project toward the machine inner side. The O-ring **21** of the second mechanical seal **7** seals (secondarily seals) between the stationary sealing ring **17** and the first case body **3**. The other configuration of the second mechanical seal **7** is the same as that of the first mechanical seal **6**, and thus the description thereof is omitted.

The sealing liquid chamber **8** is an annular space formed in the seal case **2**. A sealing liquid for lubricating and cooling sliding portions between the sealing surfaces **16a** and **17a** of the first mechanical seal **6** and sliding portions between the sealing surfaces **16a** and **17a** of the second mechanical seal **7** is introduced into the sealing liquid chamber **8**. Water, an oil, a solvent, or the like, which cause no problem even when being mixed in the sealed fluid, are used as the sealing liquid, depending on the sealed fluid.

6

In the second case body **4**, a supply path **4a** for supplying the sealing liquid from the outside to the sealing liquid chamber **8** is formed. In the first case body **3**, a discharge path **3a** for discharging the sealing liquid from the sealing liquid chamber **8** to the outside is formed. Accordingly, the sealing liquid supplied from the supply path **4a** to the sealing liquid chamber **8** during operation of the rotary machine **50** lubricates and cools each sliding portion of the first and second mechanical seals **6** and **7** and is discharged through the discharge path **3a** to the outside.

[Circulation Pump Mechanism]

FIG. 2 is a cross-sectional view as seen in the direction of arrows I-I in FIG. 1. In FIG. 1 and FIG. 2, the circulation pump mechanism **30** circulates the sealing liquid in the sealing liquid chamber **8** and is placed between the first mechanical seal **6** and the second mechanical seal **7** in the seal case **2**. The circulation pump mechanism **30** includes a rotor **31**, a plurality of vanes **32**, a plurality of springs (biasing members) **33**, a housing **34**, a plurality of pump chambers **38**, and retaining members **42**. In FIG. 2, cross-sections of the rotary shaft **51**, the vanes **32**, and the retaining members **42** are omitted.

The rotor **31** is composed of, for example, an annular member made of a metal such as stainless steel. The rotor **31** is fixed to the outer circumferential side of the sleeve **9**. Specifically, a key **39** fixed to the outer circumference of the sleeve **9** is engaged with a key groove **31a** formed on the inner circumferential side of the rotor **31**, thereby fixing the rotor **31** to the outer circumferential side of the sleeve **9**. Accordingly, the rotor **31** is mounted on the rotary shaft **51** side so as to be integrally rotatable with the rotary shaft **51**, and the rotary torque of the rotary shaft **51** is transmitted to the rotor **31** via the sleeve **9**.

[Mounting Grooves]

FIG. 3 is an enlarged cross-sectional view showing an area around 6 o'clock in FIG. 2 in the circulation pump mechanism **30**. In FIG. 2 and FIG. 3, a plurality of mounting grooves **31c** (four in FIG. 2) are formed on the rotor **31** so as to be open at an outer circumferential surface **31b** thereof and spaced at predetermined intervals in the circumferential direction. Each mounting groove **31c** is formed over the entirety in the axial direction of the rotor **31** (direction perpendicular to the drawing sheet of FIG. 3).

Each mounting groove **31c** has a first groove portion **31d** and a second groove portion **31e**. The first groove portion **31d** is formed on the bottom side (deep side) of the mounting groove **31c**. The second groove portion **31e** is formed on the opening side of the mounting groove **31c**. A groove width **W1** of the first groove portion **31d** is larger than a groove width **W2** of the second groove portion **31e**. The depth of the first groove portion **31d** is shallower than the depth of the second groove portion **31e**.

[Retaining Members]

The retaining member **42** is fitted into the first groove portion **31d** of each mounting groove **31c** from the axially outer side thereof. FIG. 4 is a perspective view showing the retaining member **42**. In FIG. 3 and FIG. 4, each retaining member **42** is composed of a rectangular parallelepiped-shaped member made of a synthetic resin. The retaining member **42** of the present embodiment is made of, for example, a fluorine resin such as polytetrafluoroethylene (PTFE). The retaining member **42** is fitted to a pair of side surfaces **31d1** of the first groove portion **31d** with a tightening allowance. Accordingly, the retaining member **42** is fixed to the first groove portion **31d**.

The retaining member **42** has an insertion hole **42a** into which the spring **33** is inserted. A plurality of insertion holes

7

42a (two in FIG. 4) are formed in the retaining member 42 so as to be aligned in the axial direction of the rotor 31. Each insertion hole 42a is formed so as to penetrate the retaining member 42 in the depth direction of the first groove portion 31d. Each insertion hole 42a communicates with the second groove portion 31e.

[Springs]

A plurality of springs 33 are provided in each mounting groove 31c. Specifically, a plurality of springs 33 (two in this case) are individually inserted into the plurality of insertion holes 42a of the retaining member 42 fixed to each first groove portion 31d. Each spring 33 is, for example, a conical coil spring made of a metal such as stainless steel. The conical coil spring 33 is inserted into the mounting groove 31c such that the large diameter end thereof is located on the deep side. The large diameter end of the conical coil spring 33 penetrates the insertion hole 42a and is in contact with the bottom surface of the first groove portion 31d.

The length in the longitudinal direction of the conical coil spring 33 in a most contracted state (state shown in FIG. 3) thereof is longer than the depth of the first groove portion 31d. Therefore, the conical coil spring 33 has an exposed portion 33a which is always exposed from the insertion hole 42a of the retaining member 42 into the second groove portion 31e. In the present embodiment, the small diameter end side of the conical coil spring 33 is the exposed portion 33a.

[Vaness]

In FIG. 2 and FIG. 3, the vane 32 is inserted into the second groove portion 31e of each mounting groove 31c. Each vane 32 is composed of, for example, a flat plate-shaped member made of carbon. The length in the axial direction (direction perpendicular to the drawing sheet of FIG. 3) of the vane 32 is substantially the same as the length in the axial direction of the mounting groove 31c. The vane 32 is slidable relative to a pair of side surfaces 31e1 of the second groove portion 31e. Accordingly, the vane 32 freely projects and retracts with respect to the outer circumferential surface 31b of the rotor 31 in the second groove portion 31e.

FIG. 5 is a perspective view showing the vane 32. In FIG. 3 and FIG. 5, a recess 32c is formed on a back surface 32b (surface located on the deep side of the second groove portion 31e) of the vane 32. A plurality of recesses 32c (two in FIG. 5) are formed on the vane 32 so as to be aligned in the axial direction of the rotor 31.

Each recess 32c is formed with a size that allows a portion on the tip (small diameter end) side of the exposed portion 33a to be inserted therinto. Accordingly, the tip end sides of the exposed portions 33a, of the plurality of conical coil springs 33, which are exposed to the vane 32 side from the retaining member 42, are individually inserted into the plurality of recesses 32c.

The tip of the exposed portion 33a inserted into each recess 32c is in contact with the bottom surface of the recess 32c. Accordingly, the plurality of conical coil springs 33 in each mounting groove 31c bias the vane 32 in a direction in which the vane 32 is caused to project from the outer circumferential surface 31b of the rotor 31. The other portion of the exposed portion 33a of each conical coil spring 33 is exposed in a space S1 of the second groove portion 31e.

The space S1 is a space formed on the deep side of the vane 32 in the second groove portion 31e. The volume of the space S1 increases and decreases as the vane 32 projects and retracts with respect to the rotor 31. Specifically, the volume of the space S1 increases when the vane 32 projects from the

8

rotor 31, and the volume of the space S1 decreases when the vane 32 retracts into the rotor 31.

[Housing]

In FIG. 1 and FIG. 2, the housing 34 of the circulation pump mechanism 30 is mounted on the inner circumferential side of the seal case 2. Specifically, the housing 34 is placed between a step surface 3b of the first case body 3 and a projection end surface 4c of the second case body 4. The housing 34 has a cam ring 35, and a first outer side plate 36 and a second outer side plate 37 placed on both sides in the axial direction of the cam ring 35. The cam ring 35, the first outer side plate 36, and the second outer side plate 37 are restricted from rotating relative to the seal case 2 by a plurality of pins 40 (three in FIG. 2) which penetrate the cam ring 35, the first outer side plate 36, and the second outer side plate 37 in the axial direction.

A plurality of springs 41 (only one in FIG. 1) are provided between the first outer side plate 36 and the second case body 4 so as to be spaced at predetermined intervals in the circumferential direction. Each spring 41 presses the first outer side plate 36 against the second case body 4 toward the machine outer side. Accordingly, the housing 34 is held between the step surface 3b of the first case body 3 and the projection end surface 4c of the second case body 4 in a state of being biased toward the machine outer side, and is movable toward the machine inner side against the biasing forces of the springs 41. Each spring 41 may be placed between the first case body 3 and the second outer side plate 37 and press the second outer side plate 37 against the first case body 3 toward the machine inner side.

The cam ring 35 is composed of, for example, an annular member made of a metal such as stainless steel. The cam ring 35 is placed between the first case body 3 and the rotor 31. The length in the axial direction of the cam ring 35 is substantially the same as the length in the axial direction of the rotor 31. An outer circumferential surface 35a of the cam ring 35 is a circumferential surface having a center at an axis C1 of the rotary shaft 51.

An inner circumferential surface 35b of the cam ring 35 is a circumferential surface having a center C2 at a position eccentric with respect to the axis C1 of the rotary shaft 51. In the present embodiment, the center C2 of the inner circumferential surface 35b of the cam ring 35 is eccentric on the upper side of FIG. 2 with respect to the axis C1 of the rotary shaft 51. Accordingly, the inner circumferential surface 35b of the cam ring 35 is eccentric with respect to the outer circumferential surface 31b of the rotor 31. Specifically, the inner circumferential surface 35b of the cam ring 35 is eccentric so as to be gradually separated from the outer circumferential surface 31b of the rotor 31 in a first angle range $\theta 1$ from 6 o'clock to 12 o'clock in FIG. 2. In addition, the inner circumferential surface 35b of the cam ring 35 is eccentric so as to be gradually closer to the outer circumferential surface 31b of the rotor 31 in a second angle range $\theta 2$ from 12 o'clock to 6 o'clock in FIG. 2.

The inner circumferential surface 35b of the cam ring 35 is a sliding surface on which a tip 32a of each vane 32 slides as the rotor 31 rotates. Each time the rotor 31 makes one rotation in the direction of an arrow E, each vane 32 alternately repeats projection movement of projecting from the mounting groove 31c and retraction movement of retracting into the mounting groove 31c while sliding on the inner circumferential surface 35b of the cam ring 35. Each vane 32 of the present embodiment moves so as to project in the first angle range $\theta 1$ and moves so as to retract in the second angle range $\theta 2$.

In the first angle range $\theta 1$, each vane **32** gradually moves so as to project by the biasing forces of the springs **33** from a state of retracting most into the mounting groove **31c** (6 o'clock state) to a state of projecting most from the mounting groove **31c** (12 o'clock state). In the second angle range $\theta 2$, each vane **32** gradually moves so as to retract against the biasing forces of the springs **33** by being pressed against the inner circumferential surface **35b** of the cam ring **35** from the state of projecting most (12 o'clock state) to the state of retracting most (6 o'clock state).

The first outer side plate **36** and the second outer side plate **37** are each composed of, for example, a disc member made of carbon, and are placed between the first case body **3** and the sleeve **9**. Both side surfaces in the axial direction of the rotor **31** slide on the side surface on the machine outer side of the first outer side plate **36** and the side surface on the machine inner side of the second outer side plate **37**, respectively.

Due to the above configuration, the plurality of pump chambers **38** (four in FIG. 2) are formed between the outer circumferential surface **31b** of the rotor **31** and the inner circumferential surface **35b** of the cam ring **35** in the interior of the housing **34** so as to be demarcated by the vanes **32**, **32** adjacent to each other in the circumferential direction. Each pump chamber **38** is a space into and from which the sealing liquid is sucked and discharged.

The volume of each pump chamber **38** changes as the rotor **31** rotates. In the present embodiment, each time the rotor **31** makes one rotation in the direction of the arrow E, the volume of the pump chamber **38** changes so as to gradually increase in the first angle range $\theta 1$, and the volume of the pump chamber **38** changes so as to gradually decrease in the second angle range $\theta 2$.

FIG. 6 is a side view of the first outer side plate **36** of the housing **34** as seen from the machine outer side. As shown in FIG. 1 and FIG. 6, a plurality of suction ports **36a** (three in FIG. 6) are formed in the first angle range $\theta 1$ of the first outer side plate **36** so as to be spaced at predetermined intervals in the circumferential direction. Accordingly, when each pump chamber **38** passes through the first angle range $\theta 1$ due to the rotation of the rotor **31**, the volume of the pump chamber **38** gradually increases, thereby causing the pressure in the pump chamber **38** to change from a positive pressure to a negative pressure. When the pressure in the pump chamber **38** becomes negative as described above, an action of sucking the sealing liquid through each suction port **36a** into the pump chamber **38** is generated. Accordingly, the sealing liquid in the sealing liquid chamber **8** is sucked through each suction port **36a** into the pump chamber **38**.

FIG. 7 is a side view of the second outer side plate **37** of the housing **34** as seen from the machine outer side. As shown in FIG. 1 and FIG. 7, a plurality of discharge ports **37a** (three in FIG. 7) are formed in the second angle range $\theta 2$ of the second outer side plate **37** so as to be spaced at predetermined intervals in the circumferential direction. Accordingly, when each pump chamber **38** passes through the second angle range $\theta 2$ due to the rotation of the rotor **31**, the volume of the pump chamber **38** gradually decreases, thereby causing the pressure in the pump chamber **38** to change from a negative pressure to a positive pressure. When the pressure in the pump chamber **38** becomes positive as described above, an action of discharging the sealing liquid through each discharge port **37a** to the outside of the pump chamber **38** is generated. Accordingly, the sealing liquid in the pump chamber **38** is discharged through each discharge port **37a** to the sealing liquid chamber **8**.

Due to the above configuration, the circulation pump mechanism **30** functions as a vane pump, and can circulate a fixed volume of the sealing liquid each time the rotor **31** makes one rotation. In FIG. 1, for ease of understanding, the positions of the suction port **36a** and the discharge port **37a** are shown so as to be shifted in the circumferential direction. [Cut Grooves]

In FIG. 2 and FIG. 3, a plurality of cut grooves **31f** are formed on the outer circumferential surface **31b** of the rotor **31** so as to be spaced in the circumferential direction. The number of cut grooves **31f** is the same as the number of vanes **32** (four). Each cut groove **31f** is formed on the outer circumferential surface **31b** of the rotor **31** at a position, on the leading side in the rotation direction of the rotor **31**, slightly further away from the tip **32a** of the vane **32**. Each cut groove **31f** is formed in a V-shape in an axial view.

Each cut groove **31f** has a first side surface **31g** and a second side surface **31h**. The first side surface **31g** is formed on the trailing side in the rotation direction of the rotor **31**. The second side surface **31h** is formed in the leading side in the rotation direction of the rotor **31**. An angle α formed between the first side surface **31g** and the second side surface **31h** is set to have a magnitude that allows a machining tool (such as a drill) to easily access the first side surface **31g** when a communication flow path **45** described later is machined in the rotor **31** with the machining tool. [Communication Flow Paths]

The circulation pump mechanism **30** further includes a plurality of communication flow paths **45** formed in the rotor **31**. The number of communication flow paths **45** is the same as the number of vanes **32** (four). Each communication flow path **45** is a flow path that provides communication between the space **S1** of the mounting groove **31c** and the pump chamber **38**. Each communication flow path **45** of the present embodiment is formed in the rotor **31** independently of the mounting groove **31c**.

FIG. 8 is an enlarged perspective view showing an area around the cut groove **31f** of the rotor **31**. In FIG. 3 and FIG. 8, each communication flow path **45** has a plurality of flow path portions **46** aligned so as to be spaced in the axial direction of the rotor **31** (up-down direction in FIG. 8). Each flow path portion **46** is composed of, for example, a hole having a circular cross-section. One end in the longitudinal direction of each flow path portion **46** is open at the first side surface (side surface) **31g** of the cut groove **31f**. The other end in the longitudinal direction of each flow path portion **46** is open at the side surface **31e1** of the second groove portion **31e** of the mounting groove **31c**. Due to the above, each flow path portion **46** of each communication flow path **45** provides communication between the space **S1** of the mounting groove **31c** and the pump chamber **38** demarcated on the leading side in the rotation direction of the rotor **31** with respect to the tip **32a** of the vane **32** inserted into the mounting groove **31c**.

In FIG. 2 and FIG. 3, when the rotor **31** rotates in the first angle range $\theta 1$, the plurality of flow path portions **46** function as supply flow paths for supplying the sealing liquid from the pump chamber **38** into the space **S1** of the mounting groove **31c**. Specifically, in the first angle range $\theta 1$, the sealing liquid is sucked through each suction port **36a** of the first outer side plate **36** into the pump chamber **38**. Accordingly, the sealing liquid in the pump chamber **38** flows from the cut groove **31f**, which is formed on the outer circumferential surface **31b** of the rotor **31** facing the pump chamber **38**, into the plurality of flow path portions **46** which are open at the first side surface **31g** of the cut groove **31f**. The sealing liquid having flowed into the plurality of flow

11

path portions 46 flows into the space S1 of the mounting groove 31c communicating with these flow path portions 46. Therefore, when the rotor 31 rotates in the first angle range 01, the pressure of the sealing liquid having flowed into the space S1 acts as a back pressure on the back surface 32b of the vane 32.

When the rotor 31 rotates in the second angle range 02, the plurality of flow path portions 46 function as discharge flow paths for discharging the sealing liquid in the space S1 to the pump chamber 38. Specifically, in the second angle range 02, the vane 32 moves so as to retract into the mounting groove 31c against the biasing forces of the conical coil springs 33 as described above. Accordingly, the sealing liquid in the space S1 passes through the plurality of flow path portions 46 and is discharged from the cut groove 31f to the pump chamber 38. At that time, dust in the space S1 is discharged together with the sealing liquid to the pump chamber 38 via the plurality of flow path portions 46, so that the plurality of flow path portions 46 also function as flow paths for cleaning the space S1.

Advantageous Effects

As described above, in the mechanical seal device 1 of the present embodiment, the space S1 on the deep side with respect to the vane 32 in each mounting groove 31c of the rotor 31 of the circulation pump mechanism 30 communicates with the pump chamber 38 demarcated on the leading side in the rotation direction of the rotor 31 with respect to the tip 32a of the vane 32, via the communication flow path 45. Therefore, the sealing liquid in the pump chamber 38 flows into the space S1 of the mounting groove 31c via the communication flow path 45. Accordingly, the pressure of the sealing liquid in the space S1 of the mounting groove 31c and the pressure of the sealing liquid in the pump chamber 38 cancel each other, so that substantially only the biasing forces of the springs 33 act on the vane 32. Therefore, even if the pressure of the sealing liquid in the pump chamber 38 acting on the tip 32a of the vane 32 increases, interference with the projection movement of the vane 32 with respect to the rotor 31 can be suppressed.

Since each communication flow path 45 has the plurality of flow path portions 46 aligned in the axial direction of the rotor 31, the sealing liquid in the pump chamber 38 can be quickly caused to flow into the space S1 of the mounting groove 31c via the plurality of flow path portions 46.

Each communication flow path 45 is formed in the rotor 31 independently of the mounting groove 31c. Therefore, when the sealing liquid flows in the communication flow path 45 from the pump chamber 38 toward the space S1 of the mounting groove 31c, the flow of the sealing liquid does not interfere with the projection movement of the vane 32. Accordingly, interference with the projection movement of the vane 32 with respect to the rotor 31 can be further inhibited. In addition, the longitudinal direction of each communication flow path 45 is closer to a tangential direction of the outer circumferential surface 31b of the rotor 31 than the longitudinal direction of each communication flow path 45 of a second embodiment described later, so that the sealing liquid in the pump chamber 38 can be efficiently caused to flow into the space S1 of the mounting groove 31c via the communication flow path 45.

Each communication flow path 45 is formed so as to be open at the first side surface 31i of the V-shaped cut groove 31f which is formed on the outer circumferential surface 31b of the rotor 31. Therefore, the communication flow path 45 can be formed more easily than in the case where the

12

communication flow path 45 is formed so as to be open at the outer circumferential surface 31b of the rotor 31.

Since each flow path portion 46 of each communication flow path 45 is composed of a hole having a circular cross-section, the communication flow path 45 can be formed even more easily.

Second Embodiment

FIG. 9 is a cross-sectional view showing a circulation pump mechanism 30 of a mechanical seal device 1 according to the second embodiment of the present disclosure. FIG. 10 is an enlarged cross-sectional view showing an area around 6 o'clock in FIG. 9 in the circulation pump mechanism 30. The circulation pump mechanism 30 of the present embodiment is different from the first embodiment in that the retaining members 42 (see FIG. 2) into which the springs 33 are inserted are not included, and each of the configurations of the mounting grooves 31c of the rotor 31 and the communication flow paths 45 is different.

In FIG. 9 and FIG. 10, each mounting groove 31c of the rotor 31 in the circulation pump mechanism 30 of the present embodiment has a constant groove width W3 over the entirety in the depth direction thereof (see also FIG. 11). Each mounting groove 31c has a first side surface 31i and a second side surface 31j facing each other. The first side surface 31i is formed on the leading side in the rotation direction of the rotor 31. The second side surface 31j is formed on the trailing side in the rotation direction of the rotor 31.

FIG. 11 is an enlarged perspective view showing an area around the mounting groove 31c of the rotor 31. In FIG. 10 and FIG. 11, a plurality of small grooves 31k (three in FIG. 11) are formed on the first side surface 31i of each mounting groove 31c so as to be aligned in the axial direction of the rotor 31. Each small groove 31k is formed to have an arc-shaped cross-section. Each small groove 31k is formed over the entirety in the depth direction of the mounting groove 31c. Each small groove 31k is formed with a size that allows a portion in the circumferential direction at the large diameter end of the conical coil spring 33 to be inserted thereinto.

The conical coil spring 33 is inserted into each mounting groove 31c such that a portion in the circumferential direction of the conical coil spring 33 is placed in each of the plurality of small grooves 31k. The vane 32 is inserted at the front of the conical coil spring 33 in each mounting groove 31c. The vane 32 is slidable relative to the first side surface 31i and the second side surface 31j of the mounting groove 31c. The small diameter end of the conical coil spring 33 is in contact with the back surface 32b of the vane 32.

Due to the above configuration, a plurality of conical coil springs 33 (three in this case) are provided in a space S2 on the deep side with respect to the vane 32 in each mounting groove 31c so as to be aligned in the axial direction. These conical coil springs 33 bias the vane 32 in a direction in which the vane 32 is caused to project from the outer circumferential surface 31b of the rotor 31. Each conical coil spring 33 of the present embodiment is inserted into the mounting groove 31c such that the large diameter end thereof is placed on the deep side, but may be inserted into the mounting groove 31c such that the small diameter end thereof is placed on the deep side.

In FIG. 9 and FIG. 10, the plurality of small grooves 31k formed in each mounting groove 31c communicate with the space S2 on the deep side of the mounting groove 31c. In addition, the plurality of small grooves 31k are open at the

13

outer circumferential surface **31b** of the rotor **31**. Accordingly, the plurality of small grooves **31k** provide communication between the space **S2** of each mounting groove **31c** and the pump chamber **38** demarcated on the leading side in the rotation direction of the rotor **31** with respect to the mounting groove **31c**. Therefore, in the present embodiment, the plurality of small grooves **31k** formed in each mounting groove **31c** function as the plurality of flow path portions **46** of the communication flow path **45**.

Each small groove **31k** has both a function of placing a portion in the circumferential direction of the conical coil spring **33** therein and a function as the flow path portion **46**, but may function only as the flow path portion **46**. In this case, each conical coil spring **33** may be sized to be inserted within the groove width **W3** of the mounting groove **31c**. The other components of the present embodiment are the same as in the first embodiment, and thus are designated by the same reference signs, and the description thereof is omitted.

As described above, in the mechanical seal device **1** of the present embodiment as well, the sealing liquid in each pump chamber **38** flows into the space **S2** of the mounting groove **31c** via the communication flow path **45**. Accordingly, the pressure of the sealing liquid in the space **S2** of the mounting groove **31c** and the pressure of the sealing liquid in the pump chamber **38** cancel each other, so that substantially only the biasing forces of the springs **33** act on the vane **32**. Therefore, even if the pressure of the sealing liquid in the pump chamber **38** acting on the tip **32a** of the vane **32** increases, interference with the projection movement of the vane **32** with respect to the rotor **31** can be suppressed.

Since each communication flow path **45** has the plurality of flow path portions **46** (small grooves **31k**) aligned in the axial direction of the rotor **31**, the sealing liquid in the pump chamber **38** can be quickly caused to flow into the space **S2** of the mounting groove **31c** via the plurality of flow path portions **46**.

Others

The mechanical seal device **1** of the present disclosure includes a double-type mechanical seal, but it is sufficient that the mechanical seal device **1** includes at least one mechanical seal such as a single-type mechanical seal. The circulation pump mechanism **30** does not have to include the retaining members **42** into which the biasing members **33** are inserted.

The number of biasing members **33** provided in each mounting groove **31c** and the number of flow path portions **46** of each communication flow path **45** are not limited to those of the above embodiments. Each biasing member **33** is not limited to a conical coil spring, and may be, for example, a cylindrical spring.

The embodiments disclosed herein are merely illustrative and not restrictive in all aspects. The scope of the present invention is defined by the scope of the claims rather than the meaning described above, and is intended to include meaning equivalent to the scope of the claims and all modifications within the scope.

REFERENCE SIGNS LIST

- 1 mechanical seal device
- 2 seal case
- 6 first mechanical seal (mechanical seal)
- 7 second mechanical seal (mechanical seal)
- 8 sealing liquid chamber

14

- 30 circulation pump mechanism
- 31 rotor
- 31b outer circumferential surface
- 31c mounting groove
- 31f cut groove
- 31g first side surface (side surface)
- 32 vane
- 32a tip
- 33 spring (biasing member)
- 34 housing
- 35b inner circumferential surface (sliding surface)
- 38 pump chamber
- 45 communication flow path
- 46 flow path portion
- 50 rotary machine
- 51 rotary shaft
- A machine inner area
- B machine outer area
- S1, S2 space

The invention claimed is:

1. A mechanical seal device comprising:
 - a seal case surrounding a rotary shaft of a rotary machine and separating a machine inner area and a machine outer area of the rotary machine from each other;
 - at least one mechanical seal placed between the seal case and the rotary shaft;
 - a sealing liquid chamber which is formed in the seal case and into which a sealing liquid for lubricating a sliding portion of the mechanical seal is introduced; and
 - a circulation pump mechanism configured to circulate the sealing liquid in the sealing liquid chamber, wherein the circulation pump mechanism includes
 - an annular rotor provided on the rotary shaft side so as to be integrally rotatable with the rotary shaft and having a mounting groove open at an outer circumferential surface thereof,
 - a vane provided in the mounting groove and configured to freely project and retract with respect to the outer circumferential surface of the rotor,
 - a biasing member provided in a space on a deep side with respect to the vane in the mounting groove and configured to bias the vane in a direction in which the vane is caused to project from the outer circumferential surface of the rotor,
 - a housing having, on an inner circumference thereof, a sliding surface which is eccentric with respect to the outer circumferential surface of the rotor and on which a tip of the vane slides as the rotor rotates, the housing being mounted to the seal case,
 - a pump chamber which is demarcated on a leading side in a rotation direction of the rotor with respect to the tip of the vane between the outer circumferential surface of the rotor and the sliding surface of the housing and into and from which the sealing liquid is sucked and discharged, and
 - a communication flow path formed in the rotor and providing communication between the space of the mounting groove and the pump chamber.
2. The mechanical seal device according to claim 1, wherein the communication flow path has a plurality of flow path portions aligned in an axial direction of the rotor.
3. The mechanical seal device according to claim 1, wherein the communication flow path is formed in the rotor independently of the mounting groove.

15

4. The mechanical seal device according to claim 3,
wherein

a cut groove having a V-shape in an axial view is formed
at a location, facing the pump chamber, on the outer
circumferential surface of the rotor, and
the communication flow path is formed so as to be open
at a side surface of the cut groove.

5

5. The mechanical seal device according to claim 3,
wherein the communication flow path has a flow path
portion composed of a hole having a circular cross-section.

10

* * * * *

16